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(54) **SYSTEMS AND METHODS FOR
IMPLEMENTING AUTOMATIC PAYER
AUTHENTICATION**

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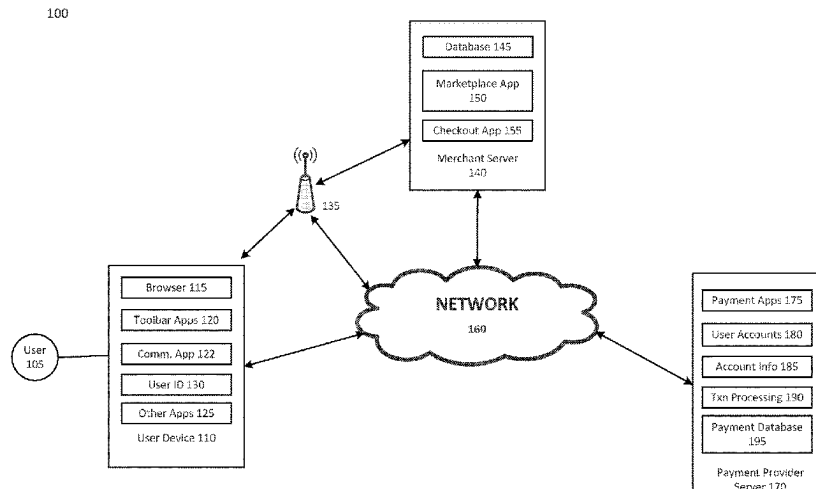
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(57) **ABSTRACT**

A system or method may be provided to facilitate automatic user authentication for electronic transactions. In particular, the system or method may automatically authenticate a customer such that the customer may make complete hands free payments without the intervention of the customer or the merchant. The automatic authentication may include a check-in process and a payment authentication process. When a customer enters a designated area of a merchant, a BLE beacon device of the merchant may automatically check in the customer at the designated area of the merchant. After the customer is checked in at the merchant's designated area, the merchant may identify the customer who is about to make a payment from a plurality of other customers who also are checked in at the merchant via Bluetooth proximity and facial recognition in parallel. Thus, the customer may automatically be authenticated to make payments by facial recognition or Bluetooth proximity.

20 Claims, 4 Drawing Sheets



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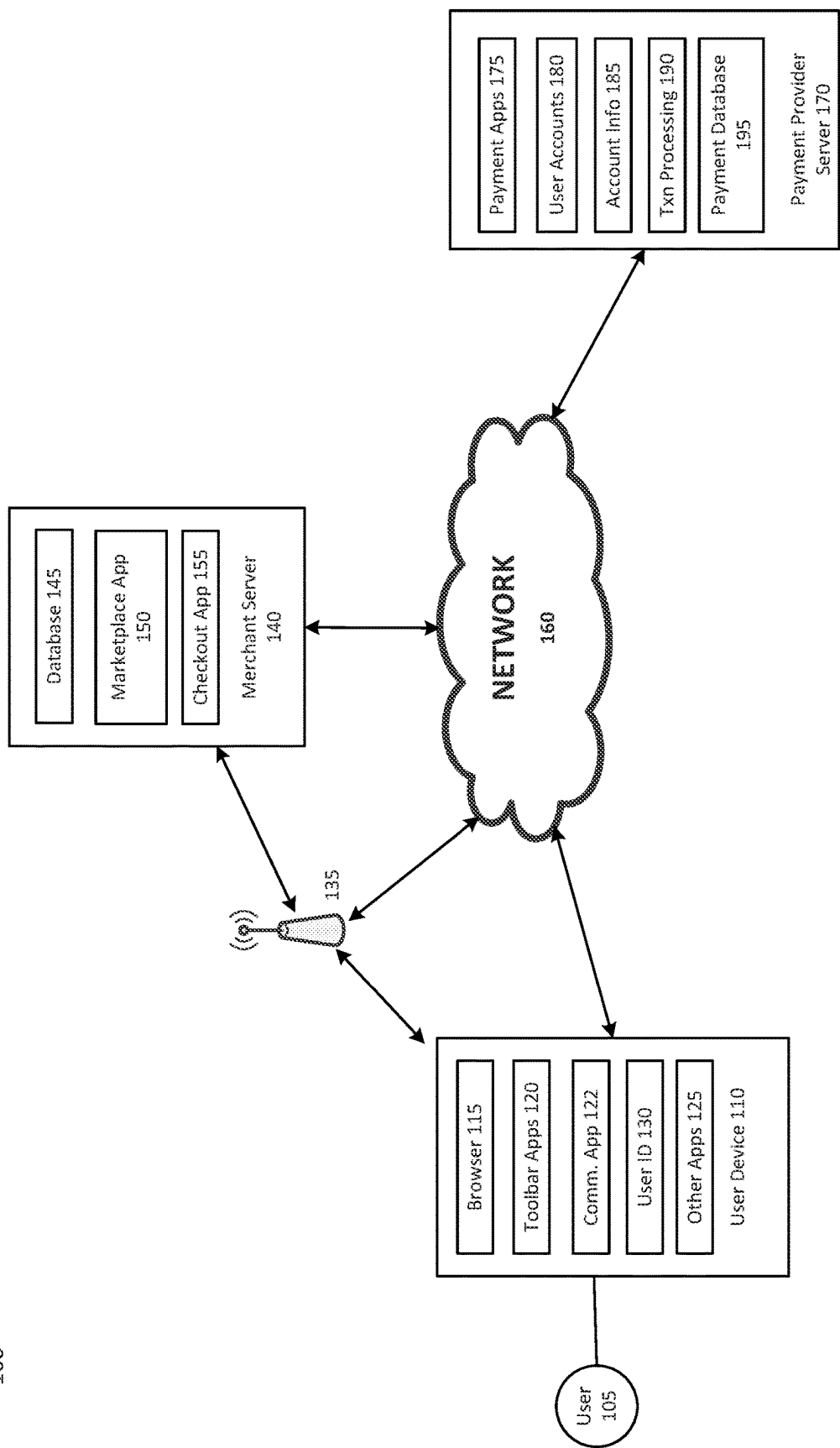


FIG. 1

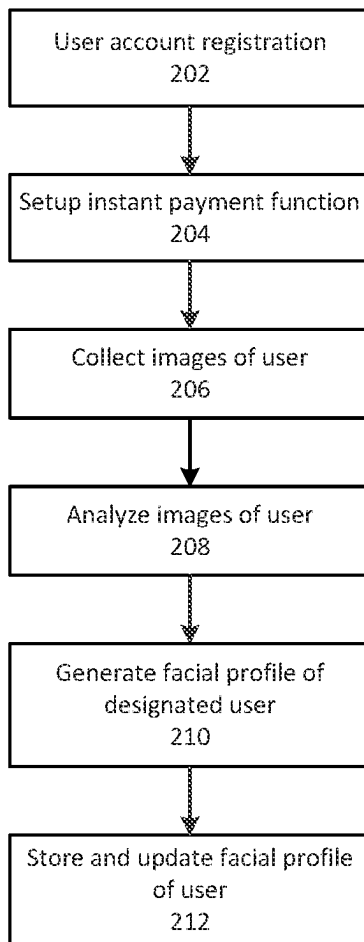


FIG. 2

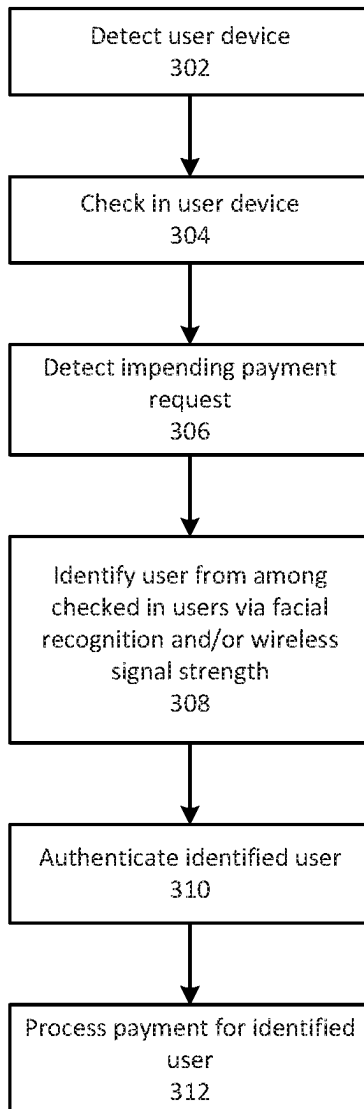
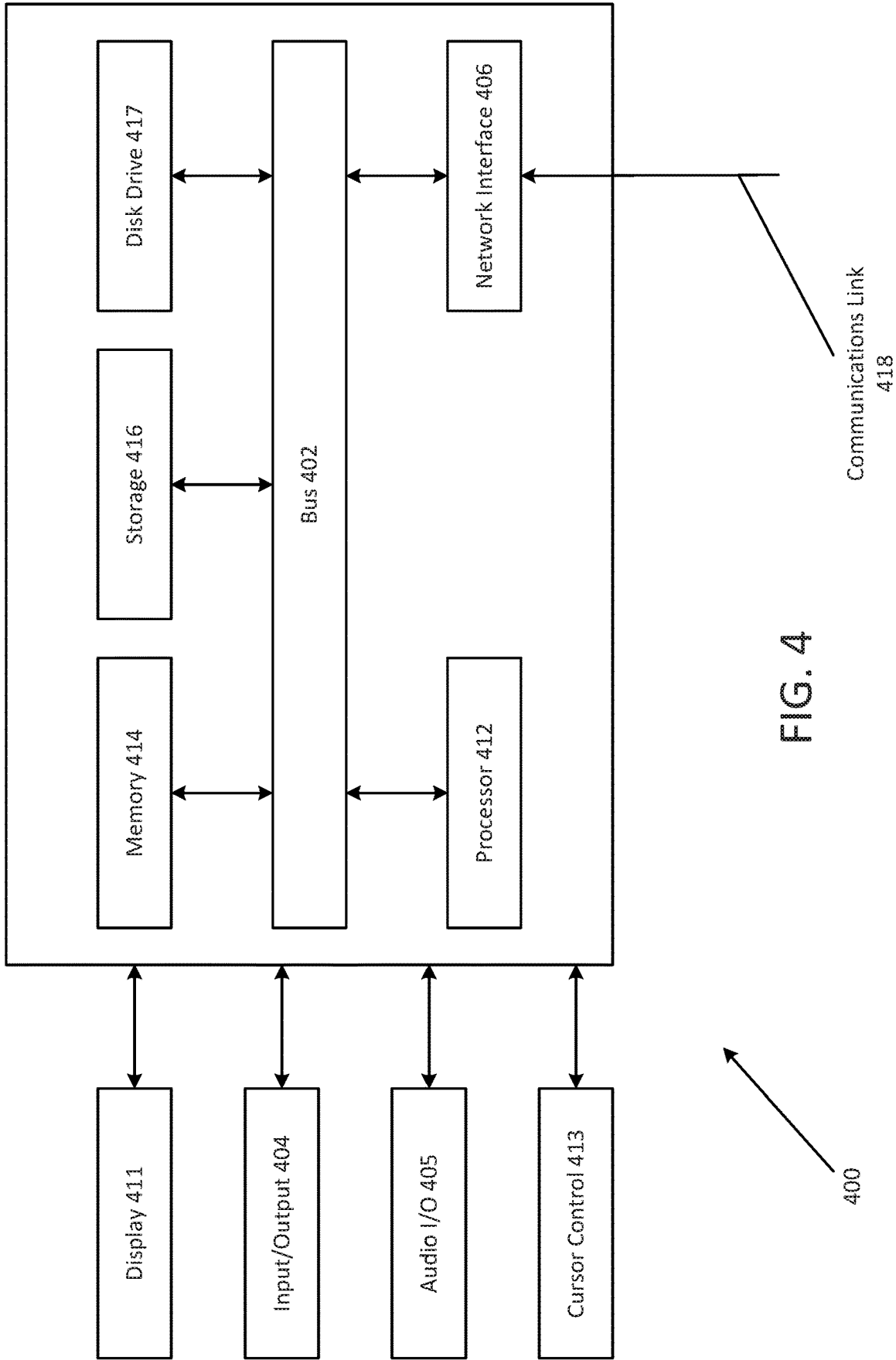


FIG. 3



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SYSTEMS AND METHODS FOR IMPLEMENTING AUTOMATIC PAYER AUTHENTICATION

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 15/882,960, filed Jan. 29, 2018, which is a continuation of U.S. patent application Ser. No. 14/297,416, filed on Jun. 5, 2014 which issued as U.S. Pat. No. 9,881,303 on Jan. 30, 2018 and whose contents of which are hereby incorporated by reference in their entirety for all purposes.

BACKGROUND

Field of the Invention

The present invention generally relates to systems and methods for implementing automatic payer authentication.

Related Art

With the popularity of electronic commerce (e-commerce), many payments are made electronically. In particular, unmanned payment stations, such as payment kiosks, gas pumps at gas stations, toll road payment booths, ticket counters, and the like, at which no personnel is present to accept payments from customers are becoming more common. A customer may use coins, cash, or credit card to make payments at these unmanned payment stations. Typically, the unmanned payment station requires certain authentication from the customer to make payments using credit cards. For example, a customer may need to swipe a credit card and enter a certain user identification (ID) and/or password to be authenticated. Thus, the authentication process may be cumbersome and inconvenient for the customer.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is block diagram of a networked system suitable for implementing automatic payer authentication according to an embodiment.

FIG. 2 is a flowchart showing a process for setting up automatic payer authentication function according to one embodiment.

FIG. 3 is a flowchart showing a process for implementing automatic payer authentication according to one embodiment.

FIG. 4 is a block diagram of a computer system suitable for implementing one or more components in FIG. 1 according to one embodiment.

Embodiments of the present disclosure and their advantages are best understood by referring to the detailed description that follows. It should be appreciated that like reference numerals are used to identify like elements illustrated in one or more of the figures, wherein showings therein are for purposes of illustrating embodiments of the present disclosure and not for purposes of limiting the same.

DETAILED DESCRIPTION

According to an embodiment, a system or method may be provided to facilitate automatic user authentication for electronic transactions. In particular, the system or method may automatically authenticate a customer such that the customer may make complete hands-free payments without the inter-

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vention of the customer or the merchant. The automatic authentication may include a check-in process and a payment authentication process.

The check-in process may be implemented when a customer enters a designated area of a merchant. The merchant may detect the customer's presence via Near Field Communication (NFC), such as Bluetooth Low Energy (BLE) communication. For example, a mobile device of the customer may be detected by a BLE beacon. The BLE beacon device of the merchant may automatically check in the customer at the designated area of the merchant when the BLE beacon device of the merchant detects the presence of the mobile device of the customer.

After the customer is properly checked in to the merchant's designated area, the merchant may identify the customer who is currently about to make a payment from among a plurality of other customers who also are checked in at the merchant. However, for unmanned payment stations, the merchant's personnel may not be present to identify the correct customer who is about to make a payment.

In an embodiment, the system or the method may implement Bluetooth proximity and facial recognition in parallel to correctly identify the customer who is about to make a payment. For example, the merchant's system may be integrated with a processor, such as a raspberry pi or an arduino board. The processor may be connected to a camera, such as a web cam, and a BLE device. When a customer is standing at the unmanned payment station at the merchant, the customer may be identified from among all the checked in customers by facial recognition. Because the system is not trying to identify the customer from the general population but from a set number of checked-in customers, the system may perform the facial recognition with relative ease. Further, by limiting the number of possible matches to the set of checked-in customers, the system may have a quick response time in the facial recognition process, and the network bandwidth consumption for implementing the facial recognition process between the merchant, the beacon, and the payment service provider may be reduced.

When a customer is checked in at the merchant location, the facial recognition service at the payment service provider may get trained on the fly for the set of checked in customers at the merchant location. When a customer walks up to the unmanned payment station of the merchant, the camera may capture the customer's image and may send the image to the payment provider server. The facial recognition service at the payment service provider may analyze the image to identify the customer from among the checked-in customers, e.g., by comparing the image against the trained data set of facial images or features. Once the customer is identified, the identification or name of the customer may be communicated to the merchant along with a confidence value indicating how closely matching the image is to the identified customer.

In an embodiment, because the facial recognition may not be 100% accurate in identifying the customer in some cases, Bluetooth proximity may be used to improve accuracy in the identification. The processor at the merchant may keep track of the Received Signal Strength Indications (RSSI) of the Bluetooth signals received from various mobile devices of the checked-in customers. The RSSIs may indicate the distances of the respective mobile devices from the unmanned payment station or from the merchant's BLE device. Thus, the RSSI values may be used to determine whether the customer identified by facial recognition is also the customer who is standing at the unmanned payment

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station. If the RSSI value and the confidence value are greater than a particular threshold, the system may confirm and authenticate the identified customer for payment. Other techniques, such as voice recognition or other biometric sensors may be used to identify the customer.

FIG. 1 is a block diagram of a networked system 100 configured to implement a process for implementing automatic payer authentication in accordance with an embodiment of the invention. Networked system 100 may comprise or implement a plurality of servers and/or software components that operate to perform various payment transactions or processes. Exemplary servers may include, for example, stand-alone and enterprise-class servers operating a server OS such as a MICROSOFT® OS, a UNIX® OS, a LINUX® OS, or other suitable server-based OS. It can be appreciated that the servers illustrated in FIG. 1 may be deployed in other ways and that the operations performed and/or the services provided by such servers may be combined or separated for a given implementation and may be performed by a greater number or fewer number of servers. One or more servers may be operated and/or maintained by the same or different entities.

System 100 may include a user device 110, a merchant server 140, a wireless beacon 135, and a payment provider server 170 in communication over a network 160. Payment provider server 170 may be maintained by a payment service provider, such as PayPal, Inc. of San Jose, CA. A user 105, such as a consumer, may utilize user device 110 to perform an electronic transaction using payment provider server 170. For example, user 105 may utilize user device 110 to visit a merchant's web site provided by merchant server 140 or the merchant's brick-and-mortar store to browse for products offered by the merchant. Further, user 105 may utilize user device 110 to initiate a payment transaction, receive a transaction approval request, or reply to the request. Note that transaction, as used herein, refers to any suitable action performed using the user device, including payments, transfer of information, display of information, etc. Although only one merchant server is shown, a plurality of merchant servers may be utilized if the user is purchasing products from multiple merchants.

User device 110, merchant server 140, wireless beacon 135, and payment provider server 170 may each include one or more processors, memories, and other appropriate components for executing instructions such as program code and/or data stored on one or more computer readable mediums to implement the various applications, data, and steps described herein. For example, such instructions may be stored in one or more computer readable media such as memories or data storage devices internal and/or external to various components of system 100, and/or accessible over network 160. Network 160 may be implemented as a single network or a combination of multiple networks. For example, in various embodiments, network 160 may include the Internet or one or more intranets, landline networks, wireless networks, and/or other appropriate types of networks.

User device 110 may be implemented using any appropriate hardware and software configured for wired and/or wireless communication over network 160. For example, in one embodiment, the user device may be implemented as a personal computer (PC), a smart phone, wearable device, laptop computer, and/or other types of computing devices capable of transmitting and/or receiving data, such as an iPad™ from Apple™.

User device 110 may include one or more browser applications 115 which may be used, for example, to provide a

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convenient interface to permit user 105 to browse information available over network 160. For example, in one embodiment, browser application 115 may be implemented as a web browser configured to view information available over the Internet, such as a user account for online shopping and/or merchant sites for viewing and purchasing goods and services. User device 110 may also include one or more toolbar applications 120 which may be used, for example, to provide client-side processing for performing desired tasks in response to operations selected by user 105. In one embodiment, toolbar application 120 may display a user interface in connection with browser application 115.

User device 110 also may include other applications to perform functions, such as email, texting, voice and IM applications that allow user 105 to send and receive emails, calls, and texts through network 160, as well as applications that enable the user to communicate, transfer information, make payments, and otherwise utilize a smart wallet through the payment provider as discussed above.

User device 110 may include one or more user identifiers 130 which may be implemented, for example, as operating system registry entries, cookies associated with browser application 115, identifiers associated with hardware of user device 110, or other appropriate identifiers, such as used for payment/user/device authentication. In one embodiment, user identifier 130 may be used by a payment service provider to associate user 105 with a particular account maintained by the payment provider. A communications application 122, with associated interfaces, enables user device 110 to communicate within system 100.

User device 110 may install and execute a payment application received from the payment service provider to facilitate payment processes. The payment application may allow user to send payment transaction requests to the payment service provider. In particular, the payment application may authenticate user 105 before making payments. In an embodiment, the payment application may implement automatic authentication of the user 105 when the user 105 is at certain payment locations.

Merchant server 140 may be maintained, for example, by a merchant or seller offering various products and/or services. The merchant may have a physical point-of-sale (POS) store front. The merchant may be a participating merchant who has a merchant account with the payment service provider. Merchant server 140 may be used for POS or online purchases and transactions. Generally, merchant server 140 may be maintained by anyone or any entity that receives money, which includes charities as well as retailers and restaurants. For example, a purchase transaction may be a donation to charity. Merchant server 140 may include a database 145 identifying available products and/or services (e.g., collectively referred to as items) which may be made available for viewing and purchase by user 105. Accordingly, merchant server 140 also may include a marketplace application 150 which may be configured to serve information over network 360 to browser 115 of user device 110. In one embodiment, user 105 may interact with marketplace application 150 through browser applications over network 160 in order to view various products, food items, or services identified in database 145.

Merchant server 140 also may include a checkout application 155 which may be configured to facilitate the purchase by user 105 of goods or services online or at a physical POS or store front. Checkout application 155 may be configured to accept payment information from or on behalf of user 105 through payment provider server 170 over network 160. For example, checkout application 155 may

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receive and process a payment confirmation from payment provider server 170, as well as transmit transaction information to the payment provider and receive information from the payment provider (e.g., a transaction ID). Checkout application 155 may be configured to receive payment via a plurality of payment methods including cash, credit cards, debit cards, checks, money orders, or the like.

Payment provider server 170 may be maintained, for example, by an online payment service provider which may provide payment between user 105 and the operator of merchant server 140. In this regard, payment provider server 170 may include one or more payment applications 175 which may be configured to interact with user device 110 and/or merchant server 140 over network 160 to facilitate the purchase of goods or services, communicate/display information, and send payments by user 105 of user device 110.

Payment provider server 170 also maintains a plurality of user accounts 180, each of which may include account information 185 associated with consumers, merchants, and funding sources, such as credit card companies. For example, account information 185 may include private financial information of users of devices such as account numbers, passwords, device identifiers, user names, phone numbers, credit card information, bank information, or other financial information which may be used to facilitate online transactions by user 105. Account information may also include user purchase history and user ratings. Advantageously, payment application 175 may be configured to interact with merchant server 140 on behalf of user 105 during a transaction with checkout application 155 to track and manage purchases made by users and which and when funding sources are used.

A transaction processing application 190, which may be part of payment application 175 or separate, may be configured to receive information from a user device and/or merchant server 140 for processing and storage in a payment database 195. Transaction processing application 190 may include one or more applications to process information from user 105 for processing an order and payment using various selected funding instruments, including for initial purchase and payment after purchase as described herein. As such, transaction processing application 190 may store details of an order from individual users, including funding source used, credit options available, etc. Payment application 175 may be further configured to determine the existence of and to manage accounts for user 105, as well as create new accounts if necessary.

In one embodiment, payment provider server 170 may receive information related to automatic authentication of the user 105. For example, a facial profile of the user 105 may be generated based on the user 105's facial features to allow facial recognition of the user 105 for automatic authentication. The user 105's account at the payment provider server 170 also may include settings and information of user 105's user device 110 and information for implementing checking in of the user device 110 at POS locations of various merchants. Merchant accounts at the payment provider server 170 also may store registration information of wireless beacons of various merchants at various POS locations.

Wireless beacon 135 may be operated by the merchant or the payment service provider. Wireless beacon 135 may broadcast wireless signals, such as Bluetooth Low Energy (BLE) signals or WiFi signals to nearby user devices. Wireless beacon 135 may be provided at a merchant's store or at other Point Of Sale (POS) locations to implement

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checking in of user devices. When the user device 110 detects the wireless signals from wireless beacon 135, user device 110 may begin a checking in process to check the user device 110 into the merchant's store or POS location. By checking in, the user device 110 may be pre-authenticated or pre-authorized to perform various transactions with the merchant.

FIG. 2 is a flowchart showing a process 200 for setting up automatic payer authentication function according to one embodiment. At step 202, payment provider server 170 may receive user 105's account registration. In particular, user 105 may set up a payment account at the payment service provider to make and receive payments. User 105 may set up funding sources, such as credit card accounts, bank accounts, and the like, to fund the payment account. User 105 also may provide various personal information for additional security checks. For example, user 105 may set up a login ID and a password for accessing the payment account. As such, user 105 may be authenticated when making a payment using the payment account by entering user 105's ID and password. Other authentication methods, such as finger print scanning, voice recognition techniques, or facial recognition techniques, also may be utilized for authentication.

At step 204, an automatic payer authentication function may be set up for user 105's payment account. For example, payment provider server 170 may inquire user 105 whether user 105 wants to utilize the automatic payer authentication function to automatically authenticate the user 105 when making payments without the user 105 having to implement the authentication process, e.g., without entering user ID and password. User 105 may agree to or deny the automatic payer authentication function in the payment application. At step 206, if user 105 agrees to use the instant payment function, payment provider server 170 may collect images of the user 105. The user 105 may provide or select a facial image or picture of the user 105 from user's digital photo album, contact list, social network account, and the like. The user 105 may select or provide multiple images or pictures of user 105's facial images. For example, with the user 105's permission, the user 105's social network accounts, contact lists, digital photo albums, and the like may be accessed to find pictures of the user 105. These pictures may previously be tagged by user 105 or others to identify the user 105 included in the pictures. Thus, pictures of the user 105 may be identified and collected. The user 105 also may use a camera included with user device 110 to capture a picture of the user 105, e.g., a "selfie." In some embodiments, other biometric information of the user 105, such as the user's voice, the user's finger print, and the like, may be used for authenticating the user 105. As such, the user 105 may provide or submit other biometric information to payment provider server 170. In an embodiment, the user 105 also may select or designate a funding source, such as a credit card account, that is to be used to make payments for the automatic authenticated transactions. The user 105 also may select the type of locations or merchants where the automatic authentication may be implemented, such as a toll booth, a retail location, an ATM, a payment kiosk, and the like.

At step 208, the user device 110 or the payment provider server 170 may analyze the images or pictures of the user 105. In particular, facial recognition algorithms may be used to perform image analysis to identify facial features or landmarks of the facial images. The relative position, size, and/or shape of the eyes, nose, cheekbones, jaw and the like may be analyzed. The skin texture of the payees also may be

analyzed. At step 210, a facial profile may be generated for the user 105. The facial profile may indicate features or landmarks that distinguish a face of the user 105 from the others. The facial profile of the user 105 may be continuously improved as more images or pictures of the user 105 become available for analysis. Thus, the system may continue to improve the facial profile of each payee to improve the accuracy of facial recognition.

At step 212, the facial profiles of the user 105 may be stored and continuously updated. For example, when more pictures or images of the user 105 are identified on the social network or are submitted by the user 105, the facial profile of the user 105 may continuously be updated to improve facial recognition of the user 105. In an embodiment, the facial profile of the user 105 may be improved by information and images provided from the user 105. The facial profile of the user 105 also may be improved by information and images provided from other users. Thus, the facial profile of the user 105 may be improved by crowd sourcing. In an embodiment, the facial profile of the user 105 may evolve over time as the look of a person may change over time. As such, pictures or information that are older than a certain time limit may no longer be incorporated in the facial profile of the user 105 to ensure that only most recent pictures or information are used for the facial profile.

By using the above process 200, a facial profile may be set up for each user of payment accounts for performing facial recognition. In particular, the system may collect pictures or images from the user 105's contact list, digital photo albums, social network accounts, photo accounts, and the like. The user 105 also may submit pictures of the user 105 by taking pictures of the user 105 using a camera. The collected pictures or images may be used to set up and improve the facial profiles.

FIG. 3 is a flowchart showing a process for implementing instant payments according to one embodiment. Initially, the user device 110 may be registered with the payment provider server 170. For example, the registration process may include signing up the user device 110 to the payment service provider, such as PayPal. This may involve installing a payment application on the user device 110, registering the user 105, and initializing the application with the registered user. At this point, the user device 110 may then be given a set of advance one-time use payment tokens and associated keys. In some embodiments, the associated keys may include a pair of symmetric keys. These user tokens may each have, for example, a user identifier, a token value, a key serial number and an AES or other crypto key, as will be readily appreciated by one of skill in the art. Such user tokens may be assigned by the payment provider server 170. Records of these assigned keys and user tokens are stored on database(s) at the payment provider server 170, such that it may be known to the payment service provider who such a token belongs to when it is put into use.

The wireless beacon 135 may also be registered with the payment provider server 170. For example, the wireless beacon 135 may be supplied with digital signatures and one-time use tokens. Each check-in and possible purchase or other transaction may then be tracked using a one-time token from both the user device 110 and a merchant beacon or beacon system that checks in the user device 110. As in the case of the user device 110 above, the keys and tokens for the beacon may also be assigned by and stored at the payment provider server 170 for later reference. After registration, the user device 110 and the wireless beacon 135 may now be ready for communications, check-ins and transactions.

The automatic authentication may include a check-in process and a payment authentication process. The check-in process may be implemented when a customer enters a designated area of a merchant. The merchant may detect the customer's presence via Near Field Communication (NFC), such as Bluetooth Low Energy (BLE) communication or LTE Direct. For example, the user device 110 of the user 105 may be detected by the wireless beacon 135. The wireless beacon 135 of the merchant may automatically check in the user device 110 at the designated area of the merchant when the wireless beacon 135 of the merchant detects the presence of the user device 110 of the user 105.

At step 302, the user device 110 may be detected. For example, a generic UUID may constantly be broadcasted from the wireless beacon 135. When the user device 110 enters the wireless broadcast range of the wireless beacon 135, this UUID is detected and verified as issued by the same payment service provider. The user device 110 and the wireless beacon 135 may then initiate communication.

At step 304, the user device 110 may then be automatically checked into the location of the merchant via the wireless beacon 135. For example, metadata about the location, a specific one-time use beacon token, and a digital signature may be sent from the wireless beacon 135 to the user device 110. The user device 110 may then certify the beacon token and verify the digital signature as being issued by the same payment service provider by using a public key previously provided from the payment service provider during registration. Assuming that the beacon token sent over is authentic, the user device 110 then may select one of its assigned one-time use user tokens.

The user device 110 then may encrypt both its user token value and the beacon token value together using the key associated with the user token, and then send this encrypted value back to the wireless beacon 135. Again, all of these communications between the user device 110 and the wireless beacon 135 may be on an unencrypted channel, as any other outside device that might be listening or noting these open communications will not know what to make of the token values without any reference table that knows where those tokens were assigned. In fact, the wireless beacon 135 may not know what to make of the combined encrypted value. Rather, the wireless beacon 135 may simply forward this value on a back channel to the payment provider server 170. The payment provider server 170 knows what to do with these values, since it has the details of where and to whom all tokens were assigned in various tables or other storage mechanisms on its database(s).

The payment provider server 170 may decrypt the combined encrypted value and verify the authenticity and ownership of both the user token and the beacon token. The payment provider server 170 may then approve of the user device 110 and provide affirmative check-in instructions back to the wireless beacon 135. It is worth noting that while the communication between the user device 110 and the wireless beacon 135 over BLE may be unencrypted, the communication between the wireless beacon 135 and the payment provider server 170 may be on a separate more protected channel. Neither the user device 110 nor any other device need to see these communication between wireless beacon 135 and the payment provider server 170 of the payment service provider. As the user device 110 is then checked in and the one time use tokens for the user device 110 and wireless beacon 135 are both committed to this checkin and any resulting transaction, one or more new tokens may then be optionally provided to the user device 110 and/or the wireless beacon 135. The payment provider

server **170** then may mark both of these tokens as used on its databases, whereupon the “handshake” checkin process or method then may be completed.

In various embodiments, many respective one-time use tokens may be stored on each of the various user devices and/or wireless beacons at any given time. Selection of a given token from the pool of possible tokens may be random, which adds some layer of protection from potential fraud or misuse. Further, the requirements that each token be used one time only, and that the combined encrypted token values be verified by the payment service provider may prevent or reduce the possibility of token replay or bit fiddling by unscrupulous persons who might otherwise try to make something of the unencrypted and open communications over BLE channels. Again, these tokens may be replenished one at a time as they are used and discarded or otherwise rendered unusable, such as by part of a check in process. Alternatively, or in addition, each user device and/or wireless beacon may also request more tokens from the payment provider server **170** independently as needed.

Other safety mechanisms to provide better security may include expiration dates on each token, as well as a requirement that the user device **110** be in constant communication with one or more wireless beacons at the merchant from checkin through any checkout and purchase or other transaction. In the event that communication is lost or dropped, then a new checkin with new tokens may be required if desired for security purposes. Further, it is also worth noting that the third party user device does not need to access the private keys of any beacon. Rather, the public keys may be adequate for the payment provider server **170** to verify and authenticate tokens for both the user device **110** and the wireless beacon **135** for checkin and later transaction. In some embodiments, there may be only one pair of public and private keys for all beacons at all merchants, with the private key being on the payment provider server **170**, and the public key being provided to the user device **110**.

After the user device **110** is checked in at the merchant’s location (POS) via the wireless beacon **135**, the payment provider server **170** or the merchant device **140** may detect any impending payment request or transaction at step **306**. Impending payments or transactions may be detected by various methods based on the type of transactions or POS. For example, the impending payment request or transaction may be detected by the user **105** operating a vending machine, a kiosk, a payment terminal, a toll machine, an Automatic Teller Machine (ATM), a payment device at a checkout counter, or the like. As such, the impending payment request or transaction may be detected when the user **105** is operating some kind of device at the merchant’s POS to select merchandise or to begin a payment processing. In an embodiment, the impending payment request for transaction may be detected when the user **105** approaches a device or an area of the merchant’s POS where payment or transaction may be implemented. For example, when the user **105** walks up to a checkout counter or when the user **105** approaches a payment kiosk, the impending payment request or transaction may be detected.

If an impending transaction or payment is detected indicating that a user is about to make a payment or a transaction, the payment provider server **170** may identify the user who is about to make the payment or the transaction. In particular, a plurality of users may be checked in at the merchant’s POS at the same time. Thus, the payment provider server **170** may determine which one of the plurality of checked-in users is the person who is about to make a payment or a transaction.

For unmanned payment stations, the merchant does not have personnel present to identify the correct customer who is about to make a payment. Thus, the payment provider server **170** may implement facial recognition to correctly identify the user who is about to make a payment. For example, the merchant’s system may be integrated with a processor, such as a raspberry pi or an arduino board. The processor may be connected to a camera, such as a web cam, and a BLE device or LTE Direct device. When a user is standing at the unmanned payment station at the merchant, the user may be identified from among all the checked in users by facial recognition. Because the system is not trying to identify the user from a large number of people of the general population, but from a set number of checked-in users, the system may perform the facial recognition with relative ease.

When the user **105** is checked in at the merchant location, the facial recognition service at the payment service provider may get trained on the fly for the set of checked in users at the merchant location. When a user walks up to the unmanned payment station of the merchant, the camera may capture the user’s image and may send the image to the payment provider server **170**. The facial recognition service at the payment service provider may analyze the image to identify the user at the unmanned payment station from among the checked-in users, e.g., by comparing the image against the trained dataset of facial images or features. Once the user at the unmanned payment station is identified, the identification or name of the user may be notified to the merchant along with a confidence value indicating how closely matched the image is to the identified user.

In an embodiment, the payment provider server **170** may determine whether the image of the user captured at the unmanned payment station matches any of the facial profiles of the checked-in users. For example, a predetermined confidence score may be set below which the facial profile of a checked-in user may be considered non-matching. If none of the facial profiles of the checked-in users has a confidence score above the predetermined confidence score, it may be determined that none of the facial profiles of the checked-in users matches the image of the user captured at the unmanned payment station. If no match is found, the user at the unmanned payment station may be required to enter additional information to be authenticated. As such, automatic authentication is not available when the user at the unmanned payment station cannot be identified. For example, the user at the unmanned payment station may not be registered to use the automatic authentication feature. Thus, the user may be required to go through the regular authentication process of swiping a credit card and entering additional information, such as user ID and/or password.

If the image of the user captured at the unmanned payment station matches one of the facial profiles of the checked-in users, the payment provider server **170** may automatically authenticate the identified user for immediate payment transaction at step **310**. The regular authentication process of swiping a credit card and/or entering password may be omitted. In particular, the payment provider server **170** may simply determine the transaction amount based on the merchandise and/or service selected by the user and automatically process the payment for the selected merchandise and/or service at step **312**. For example, payment service provider may debit a payment amount from the user **105**’s account and may credit the payment amount to the merchant’s account. Accordingly, the user **105** may make a payment to a merchant without swiping a credit card or entering additional information at an unmanned payment

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station, such as a payment kiosk, a vending machine, a toll machine, an automatic check-out counter, an ATM, and the like. Transactions include the user withdrawing cash from an ATM.

In an embodiment, the automatic authentication feature also may be implemented at a manned payment station to speed up the payment process. For example, at a checkout counter where a cashier person is taking payments, the automatic authentication feature may be implemented to speed up the payment process for the user and the cashier. For example, the user **105** may automatically be checked in and authenticated by facial recognition and/or wireless signal strength, such that the cashier does not require the user **105** to swipe a credit card or enter any information for making a payment. If the user **105** is automatically checked in and authenticated, the cashier may be notified that the user **105** has already been checked in and authenticated and that the cashier may go ahead and process the payment without requesting the user **105** to swipe a credit card or enter any additional information.

In an embodiment, multiple checked-in users may be identified at the unmanned payment station. In this case, the multiple checked-in users may be identified to the user or users at the unmanned payment station. The user or the users may then determine how transactions or payments should be made. For example, one of the multiple identified users may pay for the entire transaction. In another example, the payment may be distributed evenly or in various ways among the multiple identified users. This may allow a group of checked-in users to decide how a payment should be made as a group, such as at a restaurant or at a group activity.

Because the facial recognition may not be 100% accurate in identifying the user in some cases, Bluetooth proximity may be used to improve accuracy in the identification. The processor at the merchant may keep track of the Received Signal Strength Indications (RSSI) of the Bluetooth signals received from various mobile devices of the checked-in users. The RSSIs may indicate the distances of the respective user devices from the unmanned payment station or from the merchant's BLE device. Thus, the RSSI values may be used to determine whether the user identified by facial recognition also is the same user who is standing at the unmanned payment station. If the RSSI value and the confidence value are greater than a particular threshold, the system may confirm and authenticate the identified customer for payment.

In an embodiment, the Bluetooth proximity may be used to speed up the facial recognition process. For example, the facial recognition process may match the facial image captured at the payment station with the user located closest to the payment station first. If the facial image matches with the user located closest to the payment station, there is no need to continue comparing and matching the facial image captured at the payment station with other users located further from the payment station, because the user located closest to the payment station is very likely the person whose facial image is captured at the payment station. Thus, the facial recognition process may perform the facial image matching based on the distance the users are located from the payment station. This may speed up the facial recognition process.

In an embodiment, two or more wireless beacons may be used to detect the position of the user devices checked in at the merchant. For example, two BLE beacons may be mounted at two different positions separated by a predetermined distance at the unmanned payment station. When a user device is detected, the Time Difference of Arrival

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(TDA) of the wireless signal emitted from the user device and received at the two BLE beacons may be determined. By calculating the difference in time it took for the wireless signal of the user device to arrive at the two BLE beacons, the position of the user device may be determined. As such, the system may determine which user device is positioned at the unmanned payment station to authenticate the user of the user device.

In an embodiment, the wireless signals emitted from the user device and received at two different BLE beacons may be used to determine the Angle of Arrival (AOA) of the wireless signal. The AOA may then be used to determine the position of the user and to authenticate the user. In some embodiment, the AOA and the TDA of the wireless signal received from the user device at two or more different BLE beacons may be used in combination to determine the position of the user device and the user. Thus, by using two more of BLE beacons, the position of the user device and the user with respect to the payment station may be determined with accuracy.

In the above processes, the steps are executed at payment provider server **170**. In one embodiment, the steps may be executed at user device **110**, at merchant server **140**, or at wireless beacon **135**. In still another embodiment, the steps may be executed among payment provider server **170**, user device **110**, wireless beacon **135**, and merchant server **140** in coordination with each other. In the above embodiment, facial recognition is used to identify the customer. Other techniques, such as voice recognition or other biometric sensors may be used to identify the customer.

The following is an example of the implementation of the processes **200** and **300**:

A vending machine may be equipped with a camera configured to take pictures of a customer standing in front of the vending machine and a BLE beacon configured to communicate and check in user devices that are in proximity to the vending machine. When a customer walks up to the vending machine, the customer may automatically be checked in by the BLE beacon. Further, the vending machine may take a picture of the customer. The picture of the customer and the proximity of the customer to the vending machine both may be used to identify and authenticate the customer. Once the authentication is done, the customer may select an item at the vending machine to purchase.

The vending machine may send a payment request to the payment service provider. The payment request may include the amount of the purchase for the selected item and the details of the identified customer and/or the item to be purchased (where the item can be cash). The payment service provider may authorize the payment transaction and may send a response back to the vending machine. If the payment transaction is successful, the vending machine may dispense the item selected by the customer. Thus, the customer is able to purchase an item from the vending machine without having to go through the authentication process, such as swiping a card or enter a user ID and/or password at the vending machine.

FIG. **4** is a block diagram of a computer system **400** suitable for implementing one or more embodiments of the present disclosure. In various implementations, the user device may comprise a personal computing device (e.g., smart phone, a computing tablet, a personal computer, laptop, wearable device, Bluetooth device, key FOB, badge, etc.) capable of communicating with the network. The merchant and/or payment provider may utilize a network computing device (e.g., a network server) capable of com-

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municating with the network. It should be appreciated that each of the devices utilized by users, merchants, and payment providers may be implemented as computer system 400 in a manner as follows.

Computer system 400 includes a bus 402 or other communication mechanism for communicating information data, signals, and information between various components of computer system 400. Components include an input/output (I/O) component 404 that processes a user action, such as selecting keys from a keypad/keyboard, selecting one or more buttons or links, etc., and sends a corresponding signal to bus 402. I/O component 404 may also include an output component, such as a display 411 and a cursor control 413 (such as a keyboard, keypad, mouse, etc.). An optional audio input/output component 405 may also be included to allow a user to use voice for inputting information by converting audio signals. Audio I/O component 405 may allow the user to hear audio. A transceiver or network interface 406 transmits and receives signals between computer system 400 and other devices, such as another user device, a merchant server, or a payment provider server via network 360. In one embodiment, the transmission is wireless, although other transmission mediums and methods may also be suitable. A processor 412, which can be a microcontroller, digital signal processor (DSP), or other processing component, processes these various signals, such as for display on computer system 400 or transmission to other devices via a communication link 418. Processor 412 may also control transmission of information, such as cookies or IP addresses, to other devices.

Components of computer system 400 also include a system memory component 414 (e.g., RAM), a static storage component 416 (e.g., ROM), and/or a disk drive 417. Computer system 400 performs specific operations by processor 412 and other components by executing one or more sequences of instructions contained in system memory component 414. Logic may be encoded in a computer readable medium, which may refer to any medium that participates in providing instructions to processor 412 for execution. Such a medium may take many forms, including but not limited to, non-volatile media, volatile media, and transmission media. In various implementations, non-volatile media includes optical or magnetic disks, volatile media includes dynamic memory, such as system memory component 414, and transmission media includes coaxial cables, copper wire, and fiber optics, including wires that comprise bus 402. In one embodiment, the logic is encoded in non-transitory computer readable medium. In one example, transmission media may take the form of acoustic or light waves, such as those generated during radio wave, optical, and infrared data communications.

Some common forms of computer readable media includes, for example, floppy disk, flexible disk, hard disk, magnetic tape, any other magnetic medium, CD-ROM, any other optical medium, punch cards, paper tape, any other physical medium with patterns of holes, RAM, PROM, EEPROM, FLASH-EEPROM, any other memory chip or cartridge, or any other medium from which a computer is adapted to read.

In various embodiments of the present disclosure, execution of instruction sequences to practice the present disclosure may be performed by computer system 400. In various other embodiments of the present disclosure, a plurality of computer systems 400 coupled by communication link 418 to the network (e.g., such as a LAN, WLAN, PTSN, and/or various other wired or wireless networks, including telecommunications, mobile, and cellular phone networks) may

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perform instruction sequences to practice the present disclosure in coordination with one another.

Where applicable, various embodiments provided by the present disclosure may be implemented using hardware, software, or combinations of hardware and software. Also, where applicable, the various hardware components and/or software components set forth herein may be combined into composite components comprising software, hardware, and/or both without departing from the spirit of the present disclosure. Where applicable, the various hardware components and/or software components set forth herein may be separated into sub-components comprising software, hardware, or both without departing from the scope of the present disclosure. In addition, where applicable, it is contemplated that software components may be implemented as hardware components and vice-versa.

Software, in accordance with the present disclosure, such as program code and/or data, may be stored on one or more computer readable mediums. It is also contemplated that software identified herein may be implemented using one or more general purpose or specific purpose computers and/or computer systems, networked and/or otherwise. Where applicable, the ordering of various steps described herein may be changed, combined into composite steps, and/or separated into sub-steps to provide features described herein.

The foregoing disclosure is not intended to limit the present disclosure to the precise forms or particular fields of use disclosed. As such, it is contemplated that various alternate embodiments and/or modifications to the present disclosure, whether explicitly described or implied herein, are possible in light of the disclosure. Having thus described embodiments of the present disclosure, persons of ordinary skill in the art will recognize that changes may be made in form and detail without departing from the scope of the present disclosure. Thus, the present disclosure is limited only by the claims.

What is claimed is:

1. A system comprising:

a non-transitory memory storing instructions; and
a processor configured to execute the instructions to cause the system to perform operations comprising:

determining an indication of a desire to complete a transaction by a first device of a user with an entity system of an entity at an entity location of the entity, wherein the indication is determined based on a digital token associated with the first device that was received from the entity system based on first short-range wireless communications exchanged between the entity system and the first device;

sending, to the first device, a first one-time use token verifiable at the system using an encryption key;

issuing, to a second device associated with the entity system via the first short-range wireless communications, a second one-time use token for the transaction and the first device, wherein the second one-time use token is verifiable using the encryption key;

detecting, via the first short-range wireless communications, an encrypted value received at the second device from second short-range wireless communication between the first device and the second device, wherein the second short-range wireless communications indicate that the first device of the user is within a distance of a point-of-sale (POS) location within the entity location;

determining the first one-time use token and the second one-time use token associated with the entity system

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of the entity based on the encrypted value and a decryption key usable to decrypt encrypted values associated with the first device;

verifying the first and second one-time use tokens using the decryption key;

capturing, using a biometric sensor device of the entity system, a plurality of biometrics within the distance of the POS location;

analyzing the plurality of biometrics using a biometric recognition and a trained dataset of biometric features;

identifying a biometric associated with the user from the plurality of biometrics detected within the distance based on the analyzing;

authenticating the biometric captured at the entity system located at the entity location of the entity; and

transmitting, to the entity system, a confirmation and an authentication of the user of the first device to process the transaction at the entity location based in part on the receiving the encrypted value and the authenticating the biometric.

2. The system of claim 1, wherein the indication includes another biometric of the user from the first device while at the entity location of the entity.

3. The system of claim 1, wherein the analyzing includes: comparing the biometric associated with the user of the first device from the plurality of biometrics to the trained data set of features; and

determining that the biometric corresponds to a past biometric of the user that is associated with one or more features of the trained data set of features.

4. The system of claim 1, wherein the analyzing includes determining an identification of the user based on a confidence score.

5. The system of claim 1, wherein the analyzing is performed in response to a communication between the user of the first device and the entity system located at the entity location, and wherein the communication includes an exchange of at least one of the first one-time use token or the second one-time use token between the first device and the entity system.

6. The system of claim 1, wherein the operations further comprise:

in response to the transmitting the confirmation, processing the transaction received by the system with the indication.

7. The system of claim 1, wherein the analyzing includes comparing biometric features from the plurality of biometrics for a plurality of users at the entity location to the trained data set of features.

8. A method comprising:

determining, based on a digital token received from a first device of a user via a wireless network connection from an entity system of an entity, an indication of a desire to complete a transaction by the first device of the user with the entity system of the entity at an entity location of the entity, wherein the indication is determined based further on a digital token associated with the first device that was received from the entity system based on first short-range wireless communications exchanged between the entity system and the first device;

sending, to the first device, a first one-time use token verifiable at the system using an encryption key;

issuing, to a second device associated with the entity system based on the first short-range wireless communications, a second one-time use token for the transac-

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tion and the first device, wherein the second one-time use token is verifiable using the encryption key;

detecting, via the wireless network connection, an encrypted value received at the second device from second short-range wireless communications between the first device and the second device, wherein the second short-range wireless communications indicate that the first device of the user is within a distance of a point-of-sale (POS) location within the entity location;

determining the first one-time use token and the second one-time use token associated with the entity system of the entity based on the encrypted value and a digital key usable to decrypt encrypted values associated with the first device;

verifying the first and second one-time use tokens using the digital key;

capturing, using a biometric sensor device of the entity system, a plurality of biometrics within the distance of the POS location;

analyzing the plurality of biometrics using a biometric recognition and a trained dataset of biometric features;

identifying a first biometric associated with the user from the plurality of biometrics detected within the distance based on the analyzing;

authenticating the first biometric captured at the entity system located at the entity location of the entity; and

transmitting, via the wireless network connection to the entity system, a confirmation and an authentication of the user of the first device to process the transaction at the entity location based in part on the receiving the encrypted value and the authenticating the first biometric.

9. The method of claim 8, wherein the indication includes another biometric of the user from the first device while at the entity location of the entity.

10. The method of claim 8, wherein the analyzing includes:

comparing the first biometric associated with the user of the first device from the plurality of biometrics to the trained data set of features; and

determining that the first biometric corresponds to a past biometric of the user that is associated with one or more features of the trained data set of features.

11. The method of claim 8, wherein the analyzing includes determining an identification of the user based on a confidence score.

12. The method of claim 8, wherein the analyzing is performed in response to a communication between the user of the first device and the entity system located at the entity location, and wherein the communication includes an exchange of at least one of the digital signature, the first one-time use token, or the second one-time use token between the first device and the entity system.

13. The method of claim 8, further comprising:

in response to the transmitting the confirmation, processing the transaction received by the system with the indication.

14. The method of claim 8, wherein the analyzing includes comparing biometric features from the plurality of biometrics for a plurality of users at the entity location to the trained data set of features.

15. The method of claim 8, further comprising:

receiving a second biometric associated with another user of another device;

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analyzing the second biometric associated with the another user of the another device using the biometric recognition and the trained dataset of biometric features; and

processing, based at least on the analyzing the plurality of biometrics and the second biometric, the transaction using account information associated with the user of the first device and the another user of the another device.

16. A non-transitory computer-readable medium comprising instructions which, in response to execution by a computer system, cause the computer system to perform a method comprising:

determining, based on a digital token received from a first device of a user via a wireless network connection from an entity system of an entity, an indication of a desire to complete a transaction by the first device of the user with the entity system of the entity at an entity location of the entity, wherein the indication is determined based on a digital token associated with the first device that was received from the entity system based on first short-range wireless communications exchanged between the entity system and the first device;

sending, to the first device, a first one-time use token verifiable at the computer system using an encryption key;

issuing, to a second device associated with the entity system based on the first short-range wireless communications, a second one-time use token for the transaction and the first device, wherein the second one-time use token is verifiable using the encryption key;

detecting, via the wireless network connection, an encrypted value received at the second device from second short-range wireless communications between the first device and the second device, wherein the second short-range wireless communications indicate that the first device of the user is within a distance of a point-of-sale (POS) location within the entity location;

determining the first one-time use token and the second one-time use token associated with the entity system of the entity based on the encrypted value and a key of an

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encryption key pair usable to decrypt encrypted values associated with the first device;
verifying the first and second one-time use tokens using the key;

capturing, using a biometric sensor device of the entity system, a plurality of biometrics within the distance of the POS location;

analyzing the plurality of biometrics using a biometric recognition and a trained dataset of biometric features; identifying a biometric associated with the user from the plurality of biometrics detected within the distance based on the analyzing;

authenticating the biometric captured at the entity system located at the entity location of the entity; and

transmitting, via the wireless network connection to the entity system, a confirmation and an authentication of the user of the first device to process the transaction at the entity location based in part on the receiving the encrypted value and the analyzing the biometric.

17. The non-transitory computer-readable medium of claim **16**, wherein the indication includes another biometric of the user from the first device while at the entity location of the entity.

18. The non-transitory computer-readable medium of claim **16**, wherein the analyzing includes:

comparing the biometric associated with the user of the first device from the plurality of biometrics to the trained data set of features; and

determining that the biometric corresponds to a past biometric of the user that is associated with one or more features of the trained data set of features.

19. The non-transitory computer-readable medium of claim **16**, wherein the analyzing includes determining an identification of the user based on a confidence score.

20. The non-transitory computer-readable medium of claim **16**, wherein the analyzing is performed in response to a communication between the user of the first device and the entity system located at the entity location, and wherein the communication includes an exchange of at least one of the digital signature, the first one-time use token, or the second one-time use token between the first device and the entity system.

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